

Entity Relationship Diagram (ERD)

MMI230702 - Week 2

Problem 1.

Create an ER diagram for each of the following descriptions:

- a) Each company operates four departments, and each department belongs to one company.
- b) Each department in part (a) employs one or more employees, and each employee works for one department.
- c) Each of the employees in part (b) may or may not have one or more dependants, and each dependant belongs to one employee.
- d) Each employee in part (c) may or may not have an employment history.
- e) Represent all the ER diagrams described in (a), (b), (c), and (d) as a single ER diagram.

Problem 2

Design an E-R diagram for keeping track of the exploits of your favorite sports team. You should store the matches played, the scores in each match, the players in each match and individual player statistics for each match. Summary statistics should be modeled as derived attributes.

Problem 3

Construct an E-R diagram for a car-insurance company whose customers own one or more cars each. A car be uniquely identified by its license number or engine build number. You should store car's model and the year of manufacture in your databases. Each car has associated with it zero to any number of recorded accidents. An incident is always reported to police and has a unique report number along with the information of incident like location, date, and time.

Problem 4

Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. I think about the primary keys for each entity and important attributes.

Problem 5

A university registrar's office maintains data about the following entities:

- a) courses, including number, title, credits, syllabus, and prerequisites.
- b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom.
- c) students, including student-id, name, and program.
- d) instructors, including identification number, name, department, and title.

Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office. Document all assumptions that you make about the mapping constraints.

Problem 6

Using the following requirements specification, build an appropriate ER diagram.

A Management Information System (MIS) is required for managing the running of Driving Skills courses within the Motor Sports Company. As well as holding details of available courses, the system must maintain details of all individuals involved in such courses. In addition to customers and instructors, this will mean vehicle engineers (each vehicle will be the responsibility of a particular engineer) and administrative staff (who will be responsible for the administration associated with each particular course)

Each course has a number of lessons, (lessons may be theoretical as well as practical) with the system holding details of the lessons. Each lesson may involve customers from more than one course. Instructors teach on several courses and each course will involve at least one instructor. Details of teaching rooms (Capacity, equipment) must be maintained.

When an existing customer of the Motor-Sports Company asks to be booked onto a driving course, the customer is asked for some identification. The booking administrator checks this identification for validity. A check is made to see if the course requested by the customer has any prerequisites. Once it has been established that the customer has the required skills, the course dates are displayed on the view screen. On choosing the course and date, booking details are issued to the customer. (Dealing with new customers is outside the scope of the specification).

Problem 7

Assume you have been contracted by a university to develop a database system to keep track of student registration and accommodation records. The university courses are offered by schools. There are no limitations to how many courses a student can enroll in. The Schools are not responsible for student accommodation. The university owns a number of halls and each student is given a shared room key after enrollment. Each room has furniture attached to it.

- a) Identify the main entity types for the project.
- b) Identify the main relationship types and specify the multiplicity for each relationship.

Using your answers for (a) and (b), draw a single ER diagram to represent the data.